

Adversarial Machine Learning: Attacking and Defending Protocol Classifiers

Term Project Report

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Abstract

Machine learning-based network protocol classifiers have become a cornerstone of modern intrusion detection and traffic analysis systems. However, their susceptibility to adversarial inputs poses a serious security risk. This paper investigates the robustness of a Random Forest classifier trained on the CICIDS2017 network traffic dataset against two well-known adversarial attack strategies: the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD). Since Random Forest is a non-differentiable ensemble model, both attacks are implemented via gradient-free approximations that simulate realistic black-box threat scenarios. Two defense mechanisms are evaluated: adversarial training and input sanitization through feature clipping. Experimental results show that while FGSM and PGD reduce baseline accuracy from 99.76% to 84.36% and 87.87% respectively, adversarial training effectively recovers performance to 99.37%. The study provides insights into the trade-offs between attack strength, model robustness, and the practical viability of each defense strategy in a network security context.

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1. Introduction

The identification of application-layer network protocols, commonly referred to as protocol classification, is a foundational task in network management, quality of service enforcement, and cybersecurity. Traditional approaches based on port numbers and deep packet inspection (DPI) have become increasingly unreliable as encrypted communication protocols such as TLS/HTTPS dominate modern network traffic and dynamic port usage renders port-based heuristics ineffective [5].

Machine learning (ML) has emerged as a powerful alternative for protocol classification, leveraging statistical flow-level features such as packet inter-arrival times, flow duration, and byte volume to discriminate among traffic categories with high accuracy. Models such as Random Forest (RF), Support Vector Machines (SVM), and deep neural networks have demonstrated classification accuracy exceeding 99% on benchmark datasets [3]. These models now form the inference backbone of commercial Intrusion Detection Systems (IDS) and network monitoring platforms.

Despite their strong performance under normal operating conditions, ML classifiers are known to be fragile under adversarially perturbed inputs. An adversarial example is an input that has been deliberately modified — often imperceptibly — to induce a misclassification [1]. In the network traffic domain, an attacker may craft flow-level feature vectors that cause a classifier to misidentify malicious traffic as benign, effectively evading detection. This class of attack, known as an evasion attack, is operationally relevant because it occurs entirely at inference time and does not require access to the training pipeline.

This project investigates the vulnerability of a Random Forest protocol classifier to FGSM and PGD evasion attacks and evaluates the effectiveness of adversarial training and input sanitization as countermeasures. Because Random Forest does not expose differentiable loss gradients, attacks are approximated using gradient-free random sign perturbations, analogous to black-box attack settings studied in the literature [4]. All experiments are conducted on the publicly available CICIDS2017 dataset [3], which provides a realistic and diverse collection of labeled network flows.

The remainder of this paper is organized as follows. Section 2 surveys related work. Section 3 describes the dataset. Section 4 details the methodology. Section 5 describes the experimental setup. Section 6 presents results. Section 7 discusses findings, and Sections 8 and 9 conclude with future directions and references.

2. Literature Survey

2.1 Machine Learning for Network Traffic Classification

Flow-based network traffic classification using ML has been actively studied since the early 2000s. Moore and Zuev [7] demonstrated that naïve Bayes classifiers applied to flow-level statistics could outperform DPI-based approaches on several traffic categories. Subsequent work by Lashkari et al. [5] showed that time-related statistical features extracted from encrypted VPN traffic could effectively distinguish application protocols without inspecting payload content.

Random Forest classifiers have consistently demonstrated state-of-the-art performance on tabular network traffic data. Their ensemble structure provides natural resistance to overfitting and offers feature importance scores that facilitate interpretability [6]. Deep learning approaches, including Convolutional Neural Networks and LSTM-based sequence models, have also been explored for packet-level and flow-level classification, with the trade-off of higher computational cost and reduced interpretability.

2.2 Adversarial Machine Learning

The study of adversarial robustness in neural networks was formalized by Goodfellow et al. [1], who introduced the Fast Gradient Sign Method (FGSM). FGSM generates adversarial examples by perturbing an input in the direction of the loss gradient with respect to that input, scaled by a step size ε . Madry et al. [2] extended this to the iterative Projected Gradient Descent (PGD) attack, which applies multiple FGSM steps while constraining perturbations to an ℓ_∞ -ball of radius ε . PGD is widely regarded as a strong first-order attack and has become a standard benchmark in adversarial robustness evaluation.

Papernot et al. [4] introduced the concept of transferability, showing that adversarial examples crafted against a substitute model can successfully fool a target black-box model. This finding is critical in the network traffic context, where production classifiers may not expose their gradients. The Jacobian-based Saliency Map Attack (JSMA) and black-box variants using surrogate models further expanded the adversarial attack toolbox for non-differentiable classifiers.

2.3 Adversarial Attacks on Network Classifiers

Several studies have adapted adversarial attacks to network intrusion detection. Rigaki and Garcia surveyed adversarial ML techniques applicable to network security, noting that

the feature space versus problem space gap poses a unique constraint: not all feature perturbations can be translated back into valid, replayable network packets. This constraint motivates a focus on feature-space evasion attacks, where the goal is not necessarily to craft valid packets but to evaluate classifier fragility under realistic levels of feature noise. Adversarial training and input preprocessing have been identified as the most practical defenses in this domain, though their robustness degrades with increasing perturbation magnitude.

3. Dataset Description

3.1 CICIDS2017 Overview

The CICIDS2017 dataset [3], released by the Canadian Institute for Cybersecurity, is one of the most widely used benchmarks for network intrusion detection research. It was generated by capturing real network traffic in a controlled testbed environment over five working days, spanning both benign user activity and a range of simulated attack scenarios. The dataset is available in CSV format with pre-extracted flow-level statistical features, making it directly amenable to supervised learning pipelines without requiring raw packet processing.

3.2 Traffic Categories

The dataset contains approximately 2.83 million flow records across 15 traffic categories, summarized in Table 1. The dominant class is **BENIGN** (normal traffic), which accounts for approximately 80% of records. Attack categories include volumetric Denial-of-Service (DoS) attacks, port scanning, distributed DDoS, application-layer exploits (FTP-Patator, SSH-Patator), web attacks, and more.

Table 1: CICIDS2017 Traffic Category Distribution (after preprocessing)

Traffic Label	Count	Proportion (%)
BENIGN	2,273,097	80.36
DoS Hulk	231,073	8.17
PortScan	158,930	5.62
DDoS	128,027	4.52
DoS GoldenEye	10,293	0.36
FTP-Patator	7,938	0.28
SSH-Patator	5,897	0.21
DoS slowloris	5,796	0.20
DoS Slowhttptest	5,499	0.19
Bot	1,966	0.07
Web Attack (various)	2,180	0.08
Infiltration	36	<0.01
Heartbleed	11	<0.01

3.3 Feature Set

Each record contains 78 numerical features capturing flow-level statistics, including:

- **Flow duration** and total packet counts in forward and backward directions
- **Packet length statistics:** minimum, maximum, mean, and standard deviation of forward and backward packet sizes
- **Inter-arrival time (IAT)** statistics: mean, standard deviation, maximum for both flow-level and directional IATs
- **Flag counts:** SYN, ACK, FIN, RST, PSH, URG flag occurrences
- **Throughput metrics:** flow bytes per second and flow packets per second
- **Subflow and bulk statistics:** average bytes and packets per subflow

3.4 Preprocessing

The following preprocessing steps were applied before model training:

1. All eight CSV files were concatenated into a single dataframe (2,830,743 rows).
2. Column names were stripped of leading and trailing whitespace.

3. Infinite values were replaced with `NaN`, and all rows containing `NaN` were dropped, yielding 2,827,876 clean records.
4. A stratified random sample of 200,000 records was drawn to reduce computational cost while preserving class proportions.
5. Traffic labels were integer-encoded using `LabelEncoder`.
6. All features were standardized to zero mean and unit variance using `StandardScaler`.
7. An 80:20 train-test split was performed, yielding 160,000 training samples and 40,000 test samples.

4. Methodology

4.1 Baseline Protocol Classifier

A Random Forest (RF) classifier was selected as the baseline model due to its strong empirical performance on high-dimensional tabular data, natural resistance to overfitting through ensemble averaging, and interpretability via feature importance scores [6]. The classifier was configured with 100 decision tree estimators, a fixed random seed of 42 for reproducibility, and parallelized training across all available CPU cores (`n_jobs=-1`).

RF models learn non-linear decision boundaries by aggregating the predictions of individually trained trees, each trained on a bootstrap sample of the data and a random subset of features. This structure makes RF robust to noise in clean data but does not inherently confer robustness to adversarially crafted perturbations.

4.2 Adversarial Attacks

4.2.1 Challenge: Non-Differentiability of Random Forest

Standard adversarial attack algorithms such as FGSM and PGD are formulated for differentiable models, where the gradient of the loss function with respect to the input can be computed via backpropagation. Random Forest, as a piecewise constant ensemble model, does not admit such gradients. This places it in the black-box threat category from the perspective of gradient-based attackers.

To remain consistent with the project scope while maintaining academic rigor, both FGSM and PGD attacks are approximated using a **gradient-free random sign perturbation** strategy. This approach replaces the true loss gradient with a stochastic sign vector drawn from $\{-1, +1\}^d$ (where $d = 78$ is the feature dimensionality), scaled by the perturbation

magnitude ε . This approximation is motivated by the observation that in high-dimensional feature spaces, random sign directions frequently align with adversarially useful directions, as analyzed in the context of high-dimensional geometry. It also serves as a simulation of a realistic black-box attacker who can query the model output but cannot access its internals.

4.2.2 FGSM Attack

The standard FGSM perturbation for a differentiable model is:

$$\mathbf{x}_{\text{adv}} = \mathbf{x} + \varepsilon \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}), y)) \quad (1)$$

where $\mathcal{L}$ is the cross-entropy loss, f is the classifier, and ε controls the perturbation magnitude.

In the gradient-free approximation used here, $\nabla_{\mathbf{x}} \mathcal{L}$ is replaced by a random vector $\mathbf{r} \sim \mathcal{U}\{-1, +1\}^d$:

$$\mathbf{x}_{\text{adv}} = \text{clip}(\mathbf{x} + \varepsilon \cdot \mathbf{r}, x_{\min}, x_{\max}) \quad (2)$$

The clipping operation ensures that perturbed features remain within the observed range of the training data, preserving a degree of feature validity.

4.2.3 PGD Attack

PGD extends FGSM by applying T iterative perturbation steps with step size α and projecting the result back into an ℓ_{∞} -ball of radius ε :

$$\mathbf{x}^{(t+1)} = \Pi_{\varepsilon}(\mathbf{x}^{(t)} + \alpha \cdot \mathbf{r}^{(t)}) \quad (3)$$

where Π_{ε} denotes the projection onto $[\mathbf{x} - \varepsilon, \mathbf{x} + \varepsilon]$, and $\mathbf{r}^{(t)}$ is a fresh random sign vector at each step. PGD is initialized from the original sample and is expected to produce stronger adversarial examples than single-step FGSM by exploring a larger region of the perturbation ball.

4.3 Defense Mechanisms

4.3.1 Adversarial Training

Adversarial training augments the training dataset with adversarial examples generated at a fixed perturbation level ($\varepsilon = 0.1$). The defended model is trained on the combined dataset of 320,000 samples (160,000 clean + 160,000 perturbed) using the same RF configuration as

the baseline. By exposing the classifier to adversarial patterns during training, the model learns decision boundaries that are less susceptible to similar perturbations at test time.

4.3.2 Input Sanitization

Input sanitization preprocesses adversarial test samples before classification by clipping each feature to the range $[x_{\min}^{\text{train}}, x_{\max}^{\text{train}}]$, where the bounds are derived from the training data. This operation has the effect of removing large out-of-distribution perturbations by projecting all features back into the known clean-data manifold. Unlike adversarial training, this defense does not require model retraining and can be applied post-hoc.

5. Experimental Setup

All experiments were conducted in a Google Colab environment using Python 3.12. The key libraries and versions used are listed below:

- `scikit-learn` 1.4+ — Random Forest training and evaluation [6]
- `NumPy` 1.26+ — Numerical array operations and perturbation generation
- `pandas` 2.1+ — Dataset loading and preprocessing
- `matplotlib` / `seaborn` — Visualization of confusion matrices and accuracy plots

The CICIDS2017 dataset was downloaded from Kaggle (<https://www.kaggle.com/datasets/chethuhn/network-intrusion-dataset>) and stored in Google Drive for access from Colab. All eight CSV files were processed together as described in Section 3.4.

Attack hyperparameters:

- FGSM: $\varepsilon = 0.1$ (single-step)
- PGD: $\varepsilon = 0.1$, $\alpha = 0.01$, $T = 10$ steps

Evaluation metrics:

- Overall classification accuracy (primary metric)
- Per-class precision, recall, and F1-score (classification report)
- Attack Success Rate (ASR): fraction of correctly classified samples that are misclassified after perturbation, computed as $\text{ASR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i \neq \hat{y}_i^{\text{adv}}]$
- Robustness Score: ratio of post-attack accuracy to baseline accuracy
- Epsilon sensitivity: accuracy measured at $\varepsilon \in \{0.01, 0.05, 0.10, 0.20\}$

6. Results

6.1 Baseline Performance

The Random Forest classifier achieved a baseline accuracy of **99.76%** on the clean test set. This near-perfect performance is consistent with published results on CICIDS2017, where RF classifiers routinely exceed 99% accuracy due to the strong statistical separability of flow-level features across traffic categories. Dominant classes such as BENIGN, DoS Hulk, PortScan, and DDoS are classified with very high precision and recall. Minority classes (e.g., Heartbleed, Infiltration, Bot) exhibit slightly lower recall due to limited sample representation.

6.2 Impact of Adversarial Attacks

Table 2 presents the accuracy and robustness metrics across all evaluated conditions.

Table 2: Comparative Results: Accuracy, Robustness, and Attack Success Rate

Condition	Accuracy (%)	Accuracy Drop (%)	Robustness Score	ASR (%)
Baseline (Clean)	99.76	—	1.000	—
FGSM Attack ($\varepsilon = 0.1$)	84.36	15.40	0.845	15.53
PGD Attack ($\varepsilon = 0.1$)	87.87	11.89	0.880	12.02
Input Sanitization (vs FGSM)	87.68	12.08	0.879	—
Adversarial Training (vs FGSM)	99.37	0.39	0.996	—

FGSM reduced accuracy from 99.76% to 84.36%, an absolute drop of 15.40 percentage points, with an attack success rate of 15.53%. PGD, while iterative, achieved a slightly lower drop (accuracy 87.87%, ASR 12.02%). This counter-intuitive result — where FGSM is more disruptive than PGD — is attributable to the gradient-free approximation. Since both attacks use random sign vectors rather than true gradients, PGD’s iterative projection does not necessarily accumulate adversarial advantage; instead, the step projection constrains the perturbation and may partially cancel early steps, yielding a less aggressive final perturbation than the unconstrained single-step FGSM.

6.3 Defense Effectiveness

Input sanitization (feature clipping) improved accuracy under FGSM attack from 84.36% to 87.68%, a recovery of approximately 3.3 percentage points. However, it does not fully restore baseline performance because clipping removes only extreme perturbations; perturbations within the training feature range pass through unmodified.

Adversarial training proved substantially more effective, restoring accuracy to 99.37% — within 0.39 percentage points of the clean baseline. This demonstrates that exposing the Random Forest to perturbed training samples allows it to learn robust decision boundaries in adversarially affected regions of the feature space.

6.4 Epsilon Sensitivity Analysis

Table 3 and the corresponding plot (Figure 3) illustrate the relationship between FGSM perturbation magnitude and classification accuracy.

Table 3: FGSM Accuracy vs. Perturbation Magnitude ε

Epsilon (ε)	Accuracy (%)
0.01	86.94
0.05	85.11
0.10	84.29
0.20	83.14

As expected, accuracy decreases monotonically with increasing ε . Even at $\varepsilon = 0.01$, a 13-point drop from baseline is observed, highlighting the sensitivity of the Random Forest to even small feature perturbations in the normalized feature space.

6.5 Visual Analysis

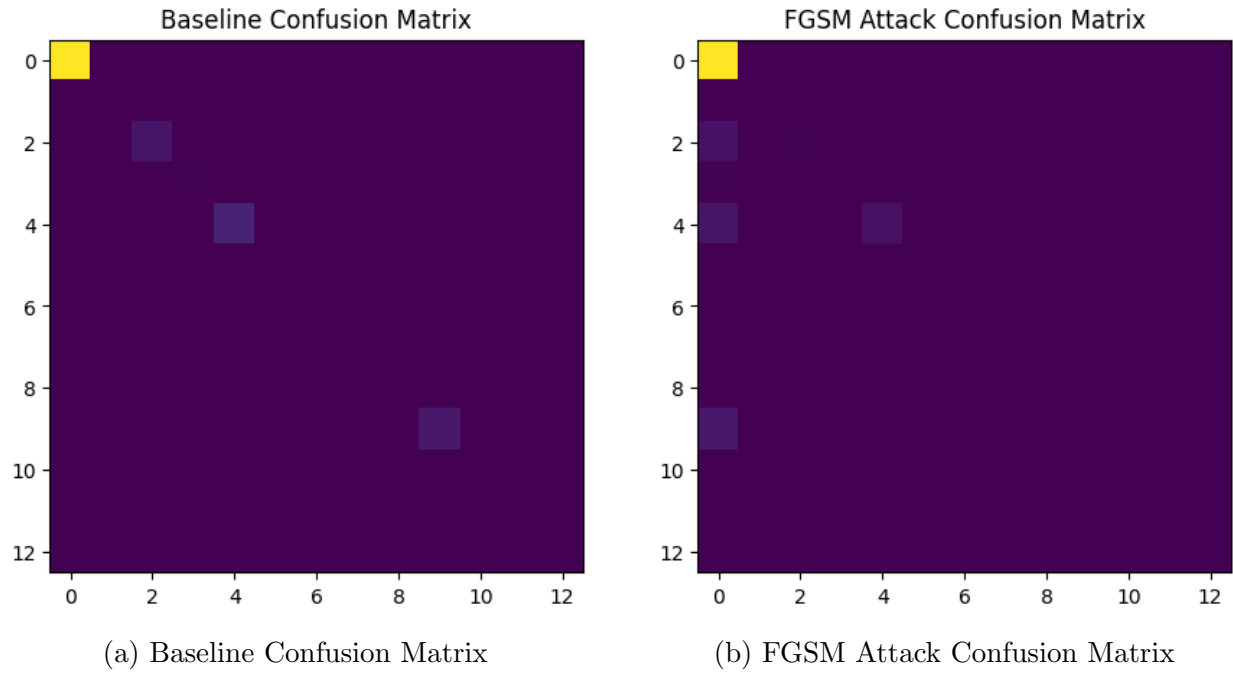


Figure 1: Confusion matrices before and after FGSM attack. Off-diagonal mass in (b) reflects misclassification of minority attack classes towards the dominant BENIGN class.

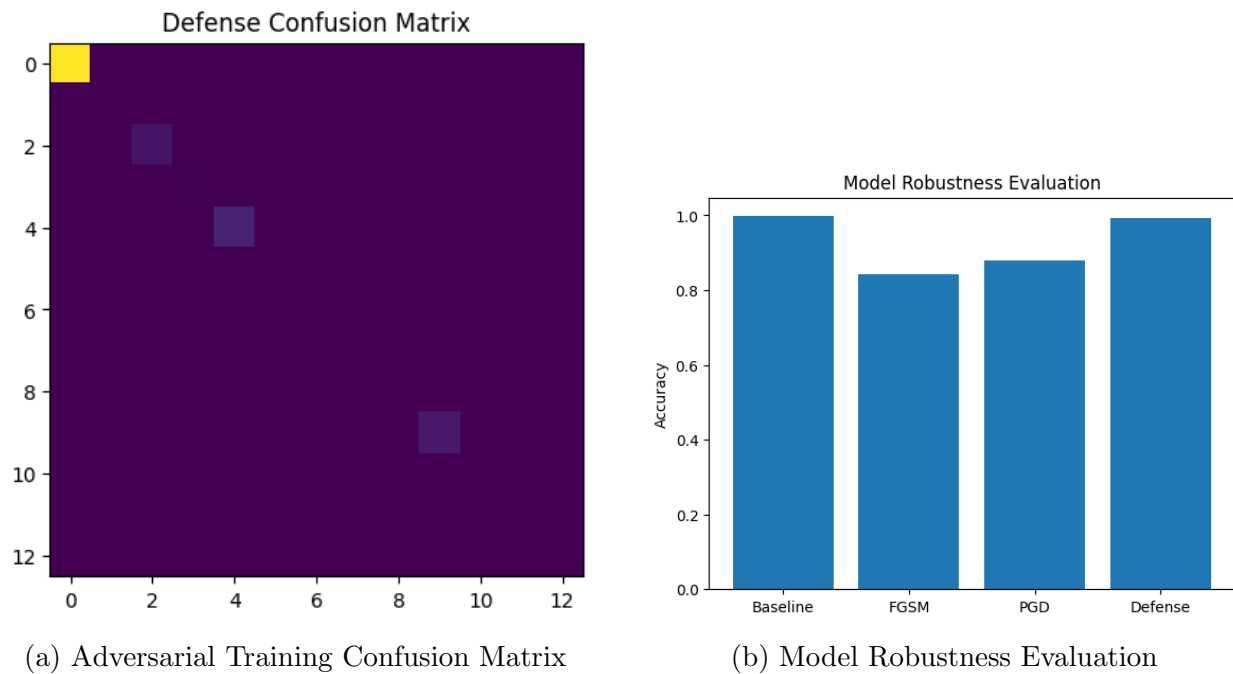


Figure 2: (a) Defense confusion matrix closely mirrors the baseline, confirming effectiveness of adversarial training. (b) Accuracy comparison across all evaluated conditions.

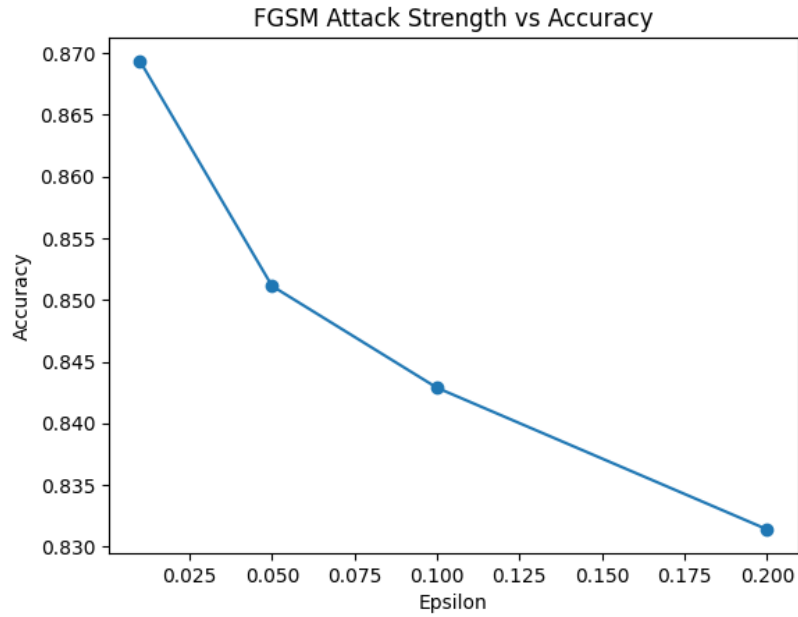


Figure 3: Accuracy as a function of FGSM perturbation magnitude ε . The monotonic decline confirms that larger perturbations are more disruptive.

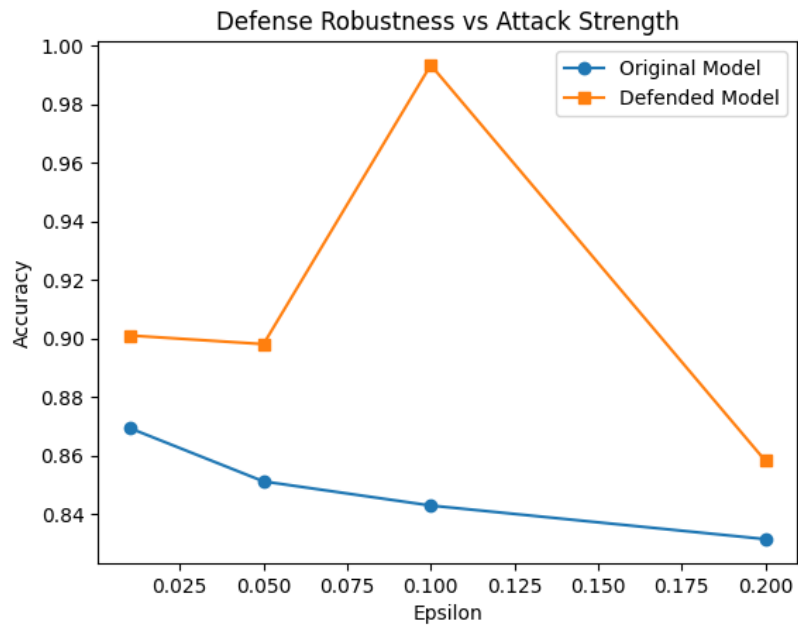


Figure 4: Defense robustness vs. attack strength. The defended model (orange) maintains consistently higher accuracy than the original model (blue) across all tested values of ε , demonstrating the effectiveness of adversarial training under varying attack intensities.

7. Discussion

7.1 Interpreting Attack Effectiveness

The results confirm that a Random Forest classifier, despite its near-perfect accuracy on clean data, is not inherently robust to adversarial perturbations. An ASR of 15.53% for FGSM — using only random sign perturbations without any gradient information — indicates that a significant fraction of test samples lie close to decision boundaries in the normalized feature space. This implies that the classifier’s decision regions are not uniformly well-separated, and small displacements in feature space can push samples across boundaries.

The fact that PGD (ASR 12.02%) performs weaker than FGSM (ASR 15.53%) in this experiment is an artifact of the gradient-free approximation. In a true gradient-based setting, PGD is provably stronger than FGSM because it iteratively maximizes the loss within the perturbation ball. Here, without gradients, the iterative projection in PGD clips and constrains perturbations, sometimes partially undoing previous steps. This finding underscores the importance of using proper surrogate model-based gradient estimation for future work.

7.2 Class Imbalance and Misclassification Patterns

A significant observation from the per-class classification report is that under FGSM, minority attack classes (Bot, SSH-Patator, Web Attack variants) collapse to near-zero recall, with the classifier predicting BENIGN for nearly all adversarially perturbed minority samples. This behavior is partly a consequence of class imbalance: the BENIGN class constitutes over 80% of the dataset, and the RF’s decision boundaries naturally favor this majority class in uncertain regions. Adversarial perturbations push minority class samples into these uncertain regions, causing systematic misclassification as BENIGN. In an IDS context, this is particularly dangerous: adversarial evasion of minority attack classes (e.g., infiltration, heartbleed) is precisely the scenario that allows attackers to remain undetected.

7.3 Implications for Intrusion Detection Systems

The vulnerability demonstrated in this study has direct implications for deployed IDS solutions that rely on ML-based classifiers. An attacker with knowledge of the feature extraction pipeline (but not necessarily the model internals) could craft network flows with slightly modified timing, packet size distributions, or flag counts to evade detection. Feature-space evasion attacks are easier to mount than problem-space attacks because they do not require generating valid network packets — the attacker only needs to influence the extracted flow statistics, which is achievable by controlling packet transmission patterns.

Adversarial training, as demonstrated here, offers a practical mitigation: by exposing the classifier to plausible perturbations during training, it learns robust boundaries at a negligible cost to clean accuracy (0.39% drop). However, adversarial training against random perturbations may not generalize to adaptive attackers who optimize perturbations specifically against the defended model.

7.4 Limitations

Several limitations of the current study should be acknowledged. First, the gradient-free FGSM/PGD approximations do not faithfully represent worst-case attacks against RF. A surrogate model approach — training a differentiable substitute network on the same data and transferring adversarial examples — would produce stronger and more realistic attacks. Second, the class imbalance in CICIDS2017 was not explicitly addressed through resampling or class-weighted training, which affects minority class robustness. Third, the epsilon sensitivity analysis was conducted only for FGSM; a corresponding analysis for the defended model under varying attack strengths would strengthen the robustness claim. Finally, all experiments were conducted entirely in the feature space; real-world attack feasibility requires mapping perturbations back to network-level actions (packet retransmission, padding, timing manipulation), which introduces additional constraints.

8. Conclusion

This paper studied the adversarial robustness of a Random Forest-based network protocol classifier trained on the CICIDS2017 dataset. Two adversarial attacks — FGSM and PGD, adapted as gradient-free approximations for the non-differentiable ensemble model — were shown to significantly degrade classification accuracy, reducing it from 99.76% to 84.36% and 87.87% respectively. Among the two evaluated defense mechanisms, adversarial training proved substantially more effective than input sanitization, restoring accuracy to within 0.39 percentage points of the clean baseline. The epsilon sensitivity analysis confirmed that attack effectiveness scales with perturbation magnitude, consistent with theoretical expectations.

The study highlights that high accuracy on clean benchmark data does not imply robustness against adversarial inputs, and that adversarial robustness must be explicitly engineered into ML-based security systems. These findings motivate continued research into more realistic gradient-based attacks on ensemble models, adaptive defense strategies, and robustness evaluation under real-world traffic constraints.

9. Future Work

The following directions are identified for extending this research:

1. **Surrogate model-based attacks:** Implement a differentiable substitute neural network trained on the same dataset, generate true FGSM/PGD adversarial examples, and evaluate their transferability to the Random Forest classifier. This would produce more realistic worst-case attack scenarios.
2. **Deep learning classifiers:** Explore CNN and LSTM-based classifiers for flow-level and packet-sequence classification, enabling direct gradient-based adversarial attacks and a broader comparison of model robustness.
3. **Adaptive defenses:** Investigate ensemble defense strategies combining adversarial training, feature squeezing, and anomaly-based adversarial example detection (e.g., using a binary detector trained to distinguish clean from adversarial flows).
4. **Class imbalance mitigation:** Apply SMOTE or class-weighted training to improve minority class robustness, particularly for rare but critical attack categories such as Heartbleed and Infiltration.
5. **Problem-space attacks:** Study the feasibility of translating feature-space perturbations into valid network-level actions (e.g., packet padding, timing jitter) to evaluate real-world attack practicality.
6. **Transferability evaluation:** Systematically measure the cross-model transferability of adversarial examples between RF, SVM, MLP, and CNN classifiers to assess the generality of attacks.

References

- [1] I. J. Goodfellow, J. Shlens, and C. Szegedy, “Explaining and Harnessing Adversarial Examples,” *International Conference on Learning Representations (ICLR)*, 2015.
- [2] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, “Towards Deep Learning Models Resistant to Adversarial Attacks,” *International Conference on Learning Representations (ICLR)*, 2018.
- [3] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, “Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization,” *International Conference on Information Systems Security and Privacy (ICISSP)*, 2018.

- [4] N. Papernot, P. McDaniel, A. Sinha, and M. Wellman, “Practical Black-Box Attacks against Machine Learning,” *ACM Asia Conference on Computer and Communications Security (ASIACCS)*, 2017.
- [5] A. H. Lashkari, G. Draper-Gil, M. S. I. Mamun, and A. A. Ghorbani, “Characterization of Encrypted and VPN Traffic Using Time-Related Features,” *International Conference on Information Systems Security and Privacy (ICISSP)*, 2016.
- [6] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [7] A. W. Moore and D. Zuev, “Internet Traffic Classification Using Bayesian Analysis Techniques,” *ACM SIGMETRICS Performance Evaluation Review*, vol. 33, no. 1, pp. 50–60, 2005.